CDA 106
Numerical Methods for Data Science

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Power Method (Iterative method)

Idea initialize $x_0 \neq 0$, $x_1 = Ax_0$, $x_2 = Ax_1$, $x_3 = Ax_2$ ---

after k iterations x_k & $x_{k+1} = Ax_k$

Then, the quotient $q = \dfrac{x_k^T A x_k}{x_k^T x_k} = \dfrac{x_k^T x_{k+1}}{x_k^T x_k} \xrightarrow{\text{approach}}$ the largest eigenvalue in the absolute sense (dominating eigenvalue)

(magnitude)

Let's consider the example

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$$

eigen values are 1, & 3

$$x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad x_1 = A x_0 = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$k = 1$

$$q = \frac{x_1^T x_2}{x_1^T x_1} = \frac{37}{13} = 2.846$$

$$x_2 = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix} \quad ; \quad x_3 = [A] \begin{bmatrix} 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 14 \\ 27 \end{bmatrix}$$

$k = 2$

$$q = \frac{x_2^T x_3}{x_2^T x_2} = \frac{[5 \ 9] \begin{bmatrix} 14 \\ 27 \end{bmatrix}}{[5 \ 9] \begin{bmatrix} 5 \\ 9 \end{bmatrix}} = \frac{313}{106} \approx 2.95 \xrightarrow{\text{approach}} \lambda = 3$$

Again consider the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$\lambda = 1, 3$

eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Let's find dominating eigen value using power method.

Sol

$x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $x_1 = Ax_0 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$x_2 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$, $x_3 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 14 \\ 13 \end{bmatrix}$

$x_4 = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 14 \\ 13 \end{bmatrix} = \begin{bmatrix} 41 \\ 40 \end{bmatrix}$

$k = 3$

$q = \dfrac{x_3^T x_4}{x_3^T x_3} = 2.997$

$$q = \frac{\begin{bmatrix} 14 & 13 \end{bmatrix} \begin{bmatrix} 41 \\ 40 \end{bmatrix}}{\begin{bmatrix} 14 & 13 \end{bmatrix} \begin{bmatrix} 14 \\ 13 \end{bmatrix}} = \frac{1094}{365} = \underline{\underline{2.997}}$$

$$\approx \boxed{\lambda = 3}$$

dominant eigen value

as soon as the quotient (q) becomes <u>closer</u> to each other at ~~consec~~ <u>consecutive</u> iterations, we should stop.

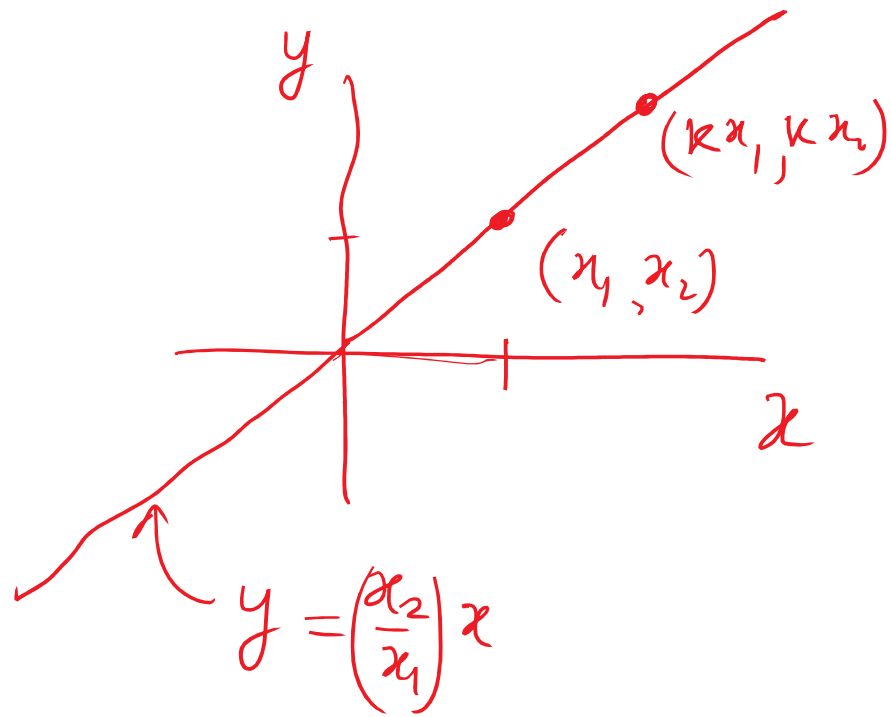
Let us now look at geometrical aspect in the last example.

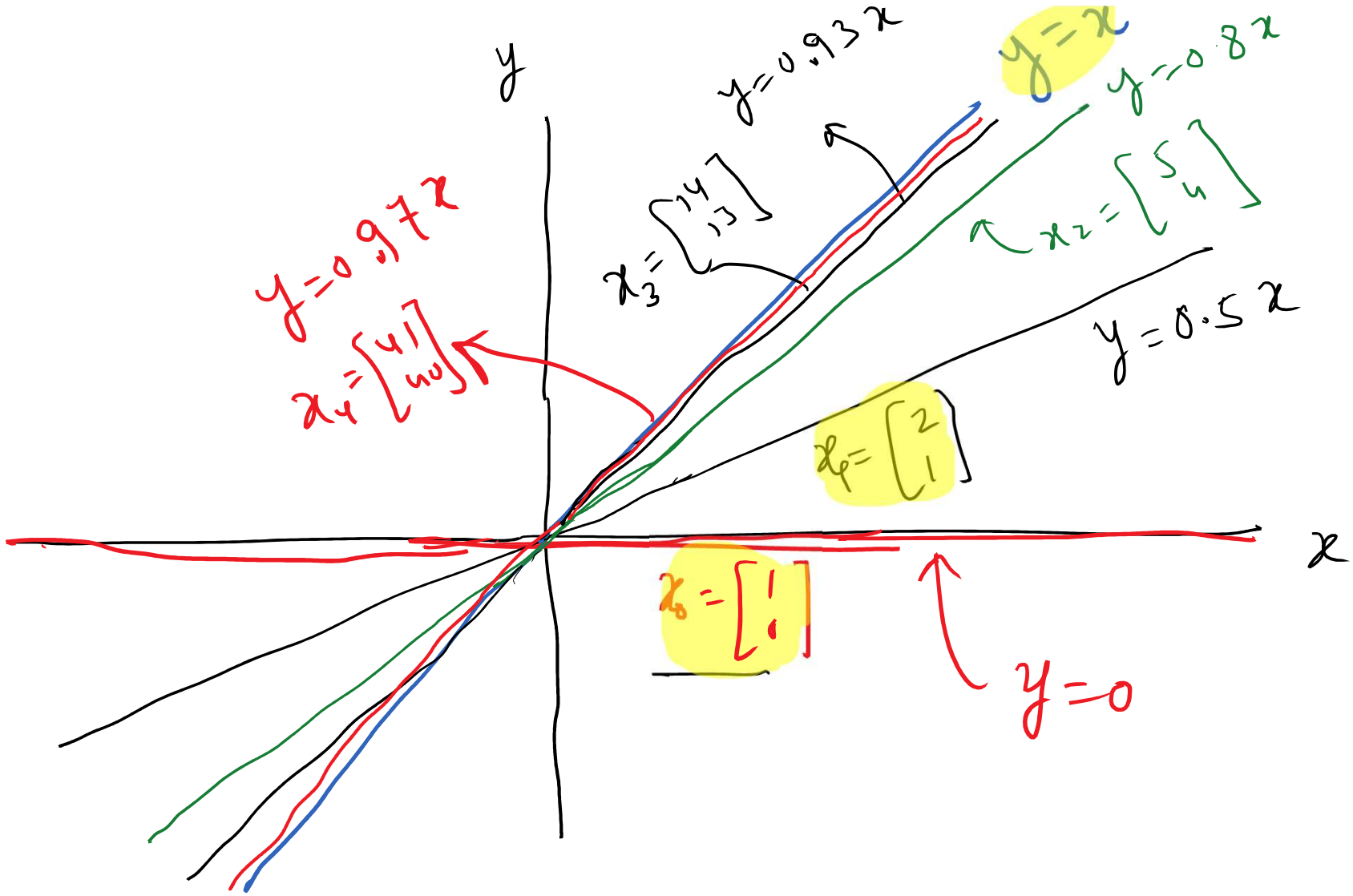
$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \qquad x = x_1 \,,\; y = x_2$$

$$x = k x_1 \,,\; y = k x_2$$

eliminate k

$$y = \left(\frac{x_2}{x_1}\right) x$$




y
x
y = 0.93x
y = x
y = 0.8x
y = 0.5x
y = 0.97x
y = 0
$x_3 = \begin{bmatrix} 14 \\ 13 \end{bmatrix}$
$x_2 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$
$x_4 = \begin{bmatrix} 41 \\ 40 \end{bmatrix}$
$x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
$x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Conclusion

(1) Power series method can be used to find dominating eigen value & corresponding eigen vector.

Consider this example for finding eigenvalue & corresponding eigen vector (using power method)

$\rightsquigarrow (\lambda_1 = 1, \lambda_2 = 3)$

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \qquad x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \; ; \quad x_2 = Ax_1 = \begin{bmatrix} 9 \\ 9 \end{bmatrix}$$

$$x_3 = Ax_2 = \begin{bmatrix} 27 \\ 27 \end{bmatrix} = 3^3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad q = \frac{x_2^T x_3}{x_2^T x_2} = \frac{\cancel{2} \times \cancel{9} \times \overset{3}{\cancel{27}}}{\cancel{2} \times \cancel{9}^{\cancel{2}}} = \underline{\underline{3}}$$

eigen vector.

eigenvalue